

Stephen Qiao

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EDUCATION

University of British Columbia

Bachelor of Applied Sciences in Electrical Engineering

Vancouver, BC

April 2026

ENGINEERING DESIGN TEAM EXPERIENCE

UBC Rover - Software-Science Integration Lead | ROS, Gazebo, Zephyr RTOS, Raspberry Pi

Aug 2023 – Present

- Developed a serial driver using Zephyr **RTOS** for the Magnetometer sensor, enabling compass navigation and detection of mineral anomalies in the rover's terrain.
- Led the design and development of firmware for a spectrophotometer used to determine iron content in unknown samples. This involved programming the device to measure light absorbance at specific wavelengths and calculate iron concentrations based on calibration curves.
- Utilized **ROS**, **Gazebo**, and **RViz** to develop, visualize, and test rover behaviors, navigation strategies, and teleoperation modules.

PROJECTS

Productivity Website | Javascript, React.js, Phaser.js, SQL, Node.js, TailwindCSS, HTML/CSS

May 2022 – Present

- Designed and launched a highly intuitive web application utilizing the Pomodoro technique, driving a user base growth of over 1,000 active users; employed **React**, **Redux**, **Node.js**, **Express**, and **SQL** for a seamless frontend and backend experience.
- Established a secure authentication system using **JWT**, **bcrypt**, and Google Sign-In, enhancing user experience and safety
- Further ensured the seamless interaction of application components with a performant **RESTful API** and adept error handling
- Created a real-time, interactive virtual garden using **Phaser.js**, transforming traditional task completion into an engaging, visually rewarding experience, thereby increasing user retention rates

Unity TSP Algorithm Visualizer | Unity, C, Shader Programming

Jul 2023 – Sep 2023

- Produced a real-time algorithm visualizer in **Unity** to solve the Traveling Salesman Problem (TSP) using various algorithms like **Brute Force**, **Nearest Neighbor**, and **Ant Colony Optimization**
- Implemented an intuitive user interface that allows users to place cities on a 2D grid, select algorithms from a dropdown menu, and visualize the algorithm's progress in real-time
- Demonstrated a deep understanding of algorithmic design by implementing multiple TSP algorithms, each with different time complexities

Gesture-Controlled Lighting System using Computer Vision | Python, OpenCV, MediaPipe

Aug 2023 – Sep 2023

- Developed an intelligent lighting control system that leverages **computer vision algorithms** to recognize hand gestures, providing a touchless interface for controlling indoor lighting.
- Implemented real-time hand gesture recognition using **MediaPipe's machine learning models**, achieving a recognition accuracy of over 95%.
- Utilized a **Raspberry Pi** and **relay module** to translate software commands into physical light control, demonstrating real-world applicability.

Magnetic Field Controlled Car | Embedded C, STM32, PIC32

Mar 2023 – May 2023

- Worked with a team of 6 to build a magnetically controlled car with a **PIC32** Timer Interrupt that measures signal pauses from a joystick controller at millisecond precision, which then converts to movement instructions
- Implemented a following mode in the car, which tracks the transmitter at up to 100 cm by detecting the strength of the magnetic field and signal strength difference between the car's left and right inductors

Reflow Oven Project | 8051 Assembly, AT89 Microcontroller, SPI Protocol, FSM Logic

Feb 2023 – Mar 2023

- Implemented advanced **FSM** logic on an **AT89LP51RC2 microcontroller** using **8051 Assembly** language, enabling real-time precision temperature control based on dynamic parameters
- Devised a temperature sensing algorithm with auditory feedback, leveraging **FSM** logic in **8051 Assembly** language to ensure consistent oven performance

- Facilitated seamless data transfer of temperature sensor readings by integrating **SPI protocol** for communication between the **AT89LP51RC2 microcontroller** and **MCP3008 ADC**

RISC-V CPU Design Project | *SystemVerilog, Quartus/Modelsim, FPGA*

Dec 2022 – Jan 2023

- Designed a versatile, **SystemVerilog-based CPU**, incorporating 12 **ARMv7** instructions spanning input-output, memory reference, and branching operations
- Crafted a comprehensive **SystemVerilog testbench** to evaluate diverse instruction combinations and validate hardware functionality
- Performed in-depth waveform analysis using **Modelsim**, successfully identifying and resolving all of discovered bugs and edge cases, ensuring robust performance

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, C#, C, C++, Verilog, Assembly, HTML, CSS, MATLAB, Bash Scripting

Hardware: Oscilloscope, Function Generator, Multimeter, Solder, FPGA, Microcontrollers

Technologies: Git, React.js, ReactNative, Redux, Unity, SQL, REST API, Linux, Node.js, Phaser.js, ARM, RTOS, ECAD, MCAD

Interests: Snowboarding, Hiking, Playing the guitar, Sketching, Cooking, Video Games